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ACCIDENT RIGHTS AGENT

Private AI for defensible claims decisions

EVIDENCE FIRST. DEADLINES VISIBLE. HUMANS REMAIN IN CONTROL.

50.9% Input token reduction	**98.8%** Repeated-case TTFT improvement
100% ProtectedField retention	**298 / 298** 30-minute stability

AMD RADEON · ROCm · LOCAL vLLM · APACHE-2.0

DataLink · Submission candidate · 2026-08-05

Track Alignment and Reading Guide

Track 2 requires core inference to run locally on AMD Radeon GPU with ROCm and does not allow a public model API for core functionality. The judging structure allocates 60 points to Agent completeness and 40 points to AMD adaptation and optimization. Accident Rights Agent implements all five listed capabilities: RAG, tool use, multi-step planning, local long-term memory, and clear permission/privacy controls.

V4 STORY	**The document and film now use one narrative.** Private deployment → Golden Case → edge cases → decisions → rights workflow → human control → measured AMD evidence → insurer pilot.

Executive Narrative

1. Problem
2. Solution
3. Innovation
4. Agent Architecture
5. Multimodal Evidence Reconstruction
6. RAG and Legal Reasoning
7. Dynamic Memory Compression
8. Rights Workflow
9. Human Approval and Safety
10. AMD ROCm Optimization
11. Performance Results
12. Deployment
13. Future Commercialization

Evidence and Limitations

EVIDENCE STANDARD	**Measured, not projected.** Radeon values come from raw JSON/CSV. Synthetic Demo results are not real-accident accuracy claims.

Executive Narrative

COMMERCIAL PROMISE	**A defensible operating record-not artificial certainty.** Faster first review, fewer evidence gaps, auditable reasoning, and controlled communication.

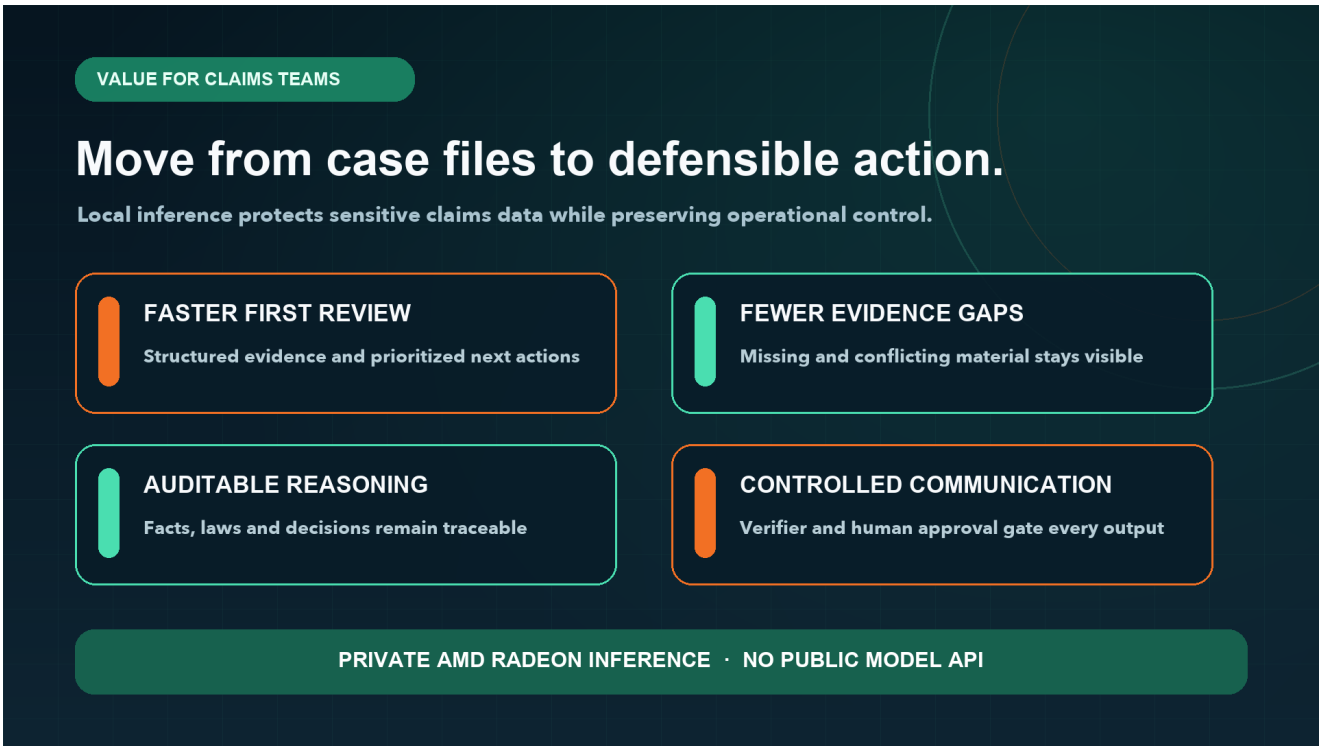


Figure 1. The insurer-value frame used in the approved V4 demonstration.

Accident Rights Agent is built around one commercial promise: preserve evidence, make deadlines visible, and keep a human claims professional in control. The approved demonstration therefore follows the same path a reviewer would use in practice: open a synthetic Golden Case, reconstruct source-linked evidence, switch to an independent conflict case, keep vehicle uncertainty and liability hypotheses visible, turn missing material into owned tasks, and stop every draft at a human approval gate.

The system does not sell certainty. It creates a defensible operating record. For an insurer, that means a faster first review, fewer evidence gaps, auditable reasoning, and controlled communication without sending sensitive case data to a public model API.

Evaluation path shown in the final demo

1. Private AMD Radeon deployment and local endpoint proof.
2. Golden Case intake, materials, Evidence Graph, and traceable timeline.
3. Conflict-case switching, vehicle parameters, and actionable missing evidence.
4. Multiple liability hypotheses and separated responsibility/payment objects.
5. Rights tasks, owners, deadlines, versioned drafts, Verifier, and Human Approval.
6. Measured Case Capsule, Prefix Cache, concurrency, and 30-minute stability results.
7. A controlled insurer pilot based on redacted cases and measurable workflow savings.

1. Problem

A traffic accident can generate dashcam video, scene photos, an accident determination, vehicle specifications, appraisal reports, service receipts, insurance documents, and later correspondence. These materials arrive at different times, may contradict one another, and are difficult to review as one case. Technical vehicle parameters can be confused with legal classification, while review or re-appraisal deadlines continue to run.

The problem affects more than vehicle owners. Claims professionals and rights-service teams need a shared, traceable case record rather than an opaque liability answer. A general-purpose chat model introduces three unacceptable risks: it may turn an unverified observation into a definitive fact, cite a non-existent or outdated rule, or merge accident liability, civil compensation, and insurance payment into one unsupported conclusion.

Accident Rights Agent focuses on a concrete scenario: after a collision between a motor vehicle and a two-wheel electric vehicle, reconstruct the evidence, preserve alternative hypotheses, identify procedural deadlines, and prepare reviewable rights-protection tasks before evidence or procedural options are lost.

The practical loss is not only legal. Claims teams repeatedly rebuild case context from messages and attachments, chase the same missing documents, and hand off conclusions without a reproducible evidence chain. A missed deadline or unsupported payment estimate can create rework, escalation, and avoidable customer distrust.

2. Solution

Accident Rights Agent is a private multimodal case workspace running on AMD Radeon GPU + ROCm. It accepts video, images, and PDFs; reconstructs an Evidence Graph and timeline; evaluates material completeness, vehicle classification, and liability hypotheses; retrieves versioned legal sources; plans evidence and deadline tasks; and produces an auditable draft for human review.

The product has two explicitly separated modes:

- mock uses deterministic adapters for functional testing without a GPU;
- local_vllm calls Qwen3-VL through the co-located loopback endpoint

http://127.0.0.1:8000/v1.

Both modes use the same CaseState, Evidence Graph, rules, RAG, memory, Verifier, Human Approval, and UI. If the local model is unavailable, the real mode stops; it never falls back to a public model API.

Actual evaluation access path

The application UI is not displayed through Radeon Cloud Jupyter or a Cloud Proxy. The evaluator opens Chrome on a Mac and forwards only the Gradio loopback port:

Mac Chrome -> SSH Tunnel (-L 7860:127.0.0.1:7860) -> Radeon Cloud Gradio -> LangGraph -> local vLLM 127.0.0.1:8000 -> AMD Radeon GPU

Port 8000 is neither forwarded nor exposed. Browser display stays on the Mac; case processing and model inference stay inside Radeon Cloud.

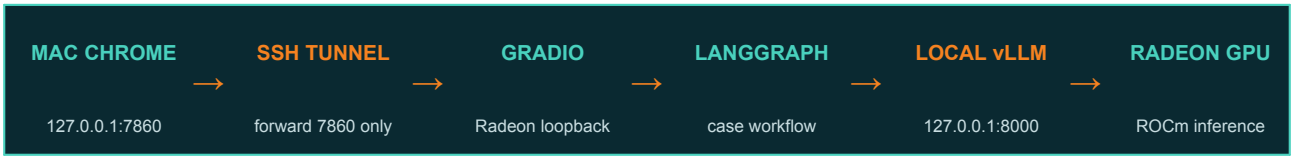


Figure 2. Approved evaluation path: the browser is local to the Mac; only Gradio 7860 is forwarded; vLLM 8000 stays inside Radeon Cloud.

3. Innovation

Evidence before conclusions

Every material fact carries a Fact ID, Evidence ID, source location, confidence, and verification state. Conflicts are recorded instead of silently overwritten. The Agent can therefore say what is known, why it is known, and what must still be checked.

Technical compliance is not legal status

The Vehicle Agent compares technical parameters but does not automatically equate an exceeded parameter with an official legal classification. It keeps four classes and multiple hypotheses when the evidence is incomplete.

Dynamic local case memory

Immutable source files, append-only Fact Memory, task-scoped Working Memory, and a verifiable Case Capsule provide long-horizon context without repeatedly sending the entire case history to the model.

Deterministic deadlines and controlled action

Deadlines are calculated by versioned rules and a workday calendar, not by model arithmetic. Documents remain drafts until a human approves the exact version and input hash. The system cannot mark an external submission as sent or accepted.

Performance and quality are measured together

AMD experiments record tokens, TTFT, TPOT, latency, throughput, GPU load, VRAM, OOM, ProtectedField retention, Verifier results, and business-quality scores. Performance gains are not accepted if protected facts or semantic quality fall below the baseline.

4. Agent Architecture

The Gradio workspace sends the active case to a LangGraph Case Orchestrator. Evidence Reconstruction parses PDFs and images, samples video frames, and creates source-linked evidence nodes. The Evidence Graph organizes facts, timeline events, conflicts, gaps, and source locations.

The business workflow then coordinates four specialized roles:

1. **Case Intake Agent** checks material completeness and routes evidence gaps.
2. **Vehicle Agent** evaluates vehicle-classification hypotheses without

collapsing technical and legal concepts.

3. **Liability Agent** builds violation, causal-force, counterfactual, and

multi-hypothesis analyses.

4. **Rights Agent** calculates deadlines, plans supplemental evidence, and

drafts review or appraisal materials.

Local Legal RAG supplies versioned LegalCitation objects. The Verifier checks evidence and legal IDs, unsupported assertions, internal consistency, risk language, and protected fields. Human Approval controls any preparation for external use. LangGraph state and retries are persisted through SQLite checkpoints.

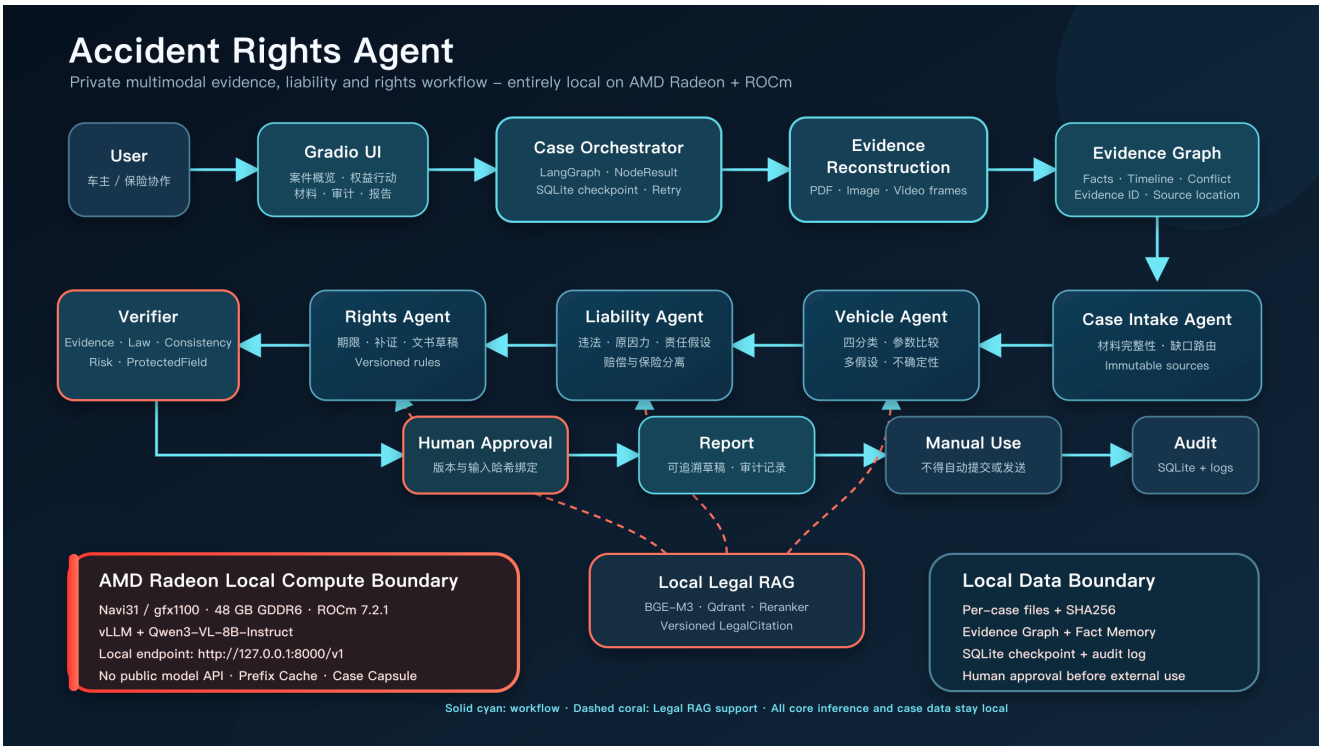


Figure 3. Evidence reconstruction, specialist Agents, local Legal RAG, Verifier, Human Approval, and the AMD Radeon compute boundary.

5. Multimodal Evidence Reconstruction

Video is not sent to the model as an unbounded stream. OpenCV samples candidate frames by interval and scene change, rejects severely blurred, dark, bright, or near-duplicate frames, keeps at most 24 frames, and sends no more than eight images in one vision request. PDFs retain page and paragraph references; scanned PDFs are rendered locally before multimodal extraction.

Each source is written atomically into its case directory with SHA256, role, MIME type, and timestamp. The intake rejects path traversal, symlinks, empty files, false extensions, and oversized files. Model-extracted facts begin as unverified and must reference an existing Evidence Node before conflict detection or human review can advance their state.

The clean Golden Case contains 12 sources, 18 facts, three timeline events, nine valid synthetic PDFs, JPEG images, and H.264 MP4 video. It contains no real personal data and does not represent real-world accident-recognition accuracy.

The independent conflict fixture is equally important: it proves that case switching resets the active view, contradictory evidence remains visible, and missing vehicle attributes become tasks instead of unsupported classifications.



Figure 4. Synthetic Golden Case UI used in the Radeon demonstration.

OPERATIONAL VALUE	**Evidence is organized before responsibility is discussed.** Source hashes, timeline events, conflicts, and missing material survive every handoff.

6. RAG and Legal Reasoning

The legal corpus stores official bibliographic identity, issuing authority, jurisdiction, version, publication and effective dates, article/paragraph/item, verbatim excerpt, source location, source URL, and content hash. The current submission contains 15 documents and 59 audited excerpts. Retrieval combines keyword and local vector results, followed by an optional local reranker. Embedding and reranking accept only local model directories.

Legal reasoning is fail-closed. A definitive legal statement without a valid LegalCitation is rejected by schema validation or the Verifier. The Liability Agent also keeps three outputs separate: accident liability, civil compensation allocation, and insurance payment estimation. If policy terms, limits, losses, or required documents are incomplete, insurance payment remains insufficient_data.

7. Dynamic Memory Compression

The baseline sends the full case history, facts, citations, and workflow state to the model. The optimized path sends a Case Capsule, task-relevant facts, required Evidence/Law IDs, protected fields, and the current workflow state. Raw evidence is never lossily compressed.

ProtectedField covers dates, deadlines, amounts, liability ratios, vehicle identifiers, party roles, Fact/Evidence/Law IDs, legal versions, and approval status. The workflow performs a set-consistency check before and after context selection and blocks the node if any protected value is lost.

On the Radeon comparison workload, input fell from 11,808 to 5,795 tokens, a 50.9% reduction, while ProtectedField retention remained 100%. Combined with Prefix Cache, TTFT p50 fell from 9.003 seconds to 0.113 seconds. These results apply to the fixed Golden Case workload and are not presented as universal ratios for every case.

8. Rights Workflow

The Rights Workflow performs service-date identification, deterministic deadline calculation, task planning, evidence-gap planning, and document draft generation. Date confidence prioritizes human confirmation, service receipts, signed or electronic delivery records, and document dates. Conflicting credible dates remain disputed.

Review and re-appraisal deadlines use versioned national rule packages and the 2026 workday calendar. The workflow can prepare tasks for vehicle-property inspection, re-inspection, supplemental appraisal, or litigation appraisal. Operational suggestions such as retaining an in-person stamped copy or using EMS are presented as traceability options, never as invented statutory requirements.

For claims operations, the output is deliberately structured as **evidence, owner, due date, next action, and draft status**. This turns analysis into a reviewable plan while leaving the insurer's existing roles and approval gates in place.



Figure 5. The UI turns missing vehicle material into a reviewable action list.

9. Human Approval and Safety

The Policy Gate aggregates evidence citation, legal citation, internal consistency, risk-language, and protected-field checks. Passing the gate means only that a draft may enter human review. Approval is bound to the document version and input hash; a changed fact or draft invalidates the old approval.

The system rejects language such as guaranteed victory, certain legal vehicle status, or inevitable primary liability. It also blocks advice to hide, delete, modify, or fabricate evidence. The output is an information-organization and decision-support artifact, not a final determination by an authority or legal opinion.

Accident liability, civil allocation, and insurance payment remain three separate objects. Missing policy terms or loss inputs keep payment at insufficient_data; they never produce a fabricated amount.

FAIL CLOSED	**Three outcomes stay separate.** Accident liability, civil allocation, and insurance payment never collapse into one unsupported answer.

10. AMD ROCm Optimization

The measured environment uses a Navi31 gfx1100 Radeon GPU with 48 GB GDDR6, ROCm 7.2.1, PyTorch ROCm 2.9.1, vLLM 0.16.1.dev0, and Qwen3-VL-8B-Instruct. Fair comparisons keep the GPU, model, Golden Case, request order, temperature=0, and output limit constant.

The baseline uses full history, Prefix Cache off, and default batch behavior. The optimized path uses Case Capsule, required fields only, and Prefix Cache. The parameter matrix covers gpu_memory_utilization 0.80/0.85/0.90/0.95, batch tokens 8192/16384/32768, and 16K/32K context. The final interactive configuration is 0.95 GPU memory utilization, 8,192 batch tokens, 16K context, and Prefix Cache enabled.

11. Performance Results

Metric	Baseline	Optimized
Input tokens	11,808	5,795
TTFT p50 / p95	9.003 / 9.007 s	0.113 / 0.113 s
TPOT p50	0.040 s/token	0.039 s/token
Latency p95	23.407 s	15.228 s
Output throughput	15.417 tok/s	25.689 tok/s
Mean GPU utilization	99.533%	100.000%
Peak VRAM	44,246 MB	43,984 MB
Success / OOM	100% / 0	100% / 0
Quality score	93.571	96.364
ProtectedField retention	100%	100%

Concurrency 1, 2, and 4 each executed 20 semantic requests. Throughput was 25.667, 49.323, and 89.478 tok/s; every group completed 20/20 requests with 100% business Verifier pass rate and zero OOM events.

The stability run lasted 1,809.493 seconds and completed 298/298 requests with zero OOM. The vLLM PID, UI/API, and SQLite quick_check remained stable. Mean GPU utilization was 99.626%, peak VRAM was 46,744 MB, and peak hotspot temperature was 61 C.

The fixed short-output performance and stability workloads do not contain the full business ID list, so their Verifier value is not used as a semantic quality claim. Full semantic concurrency results provide the business-quality gate.

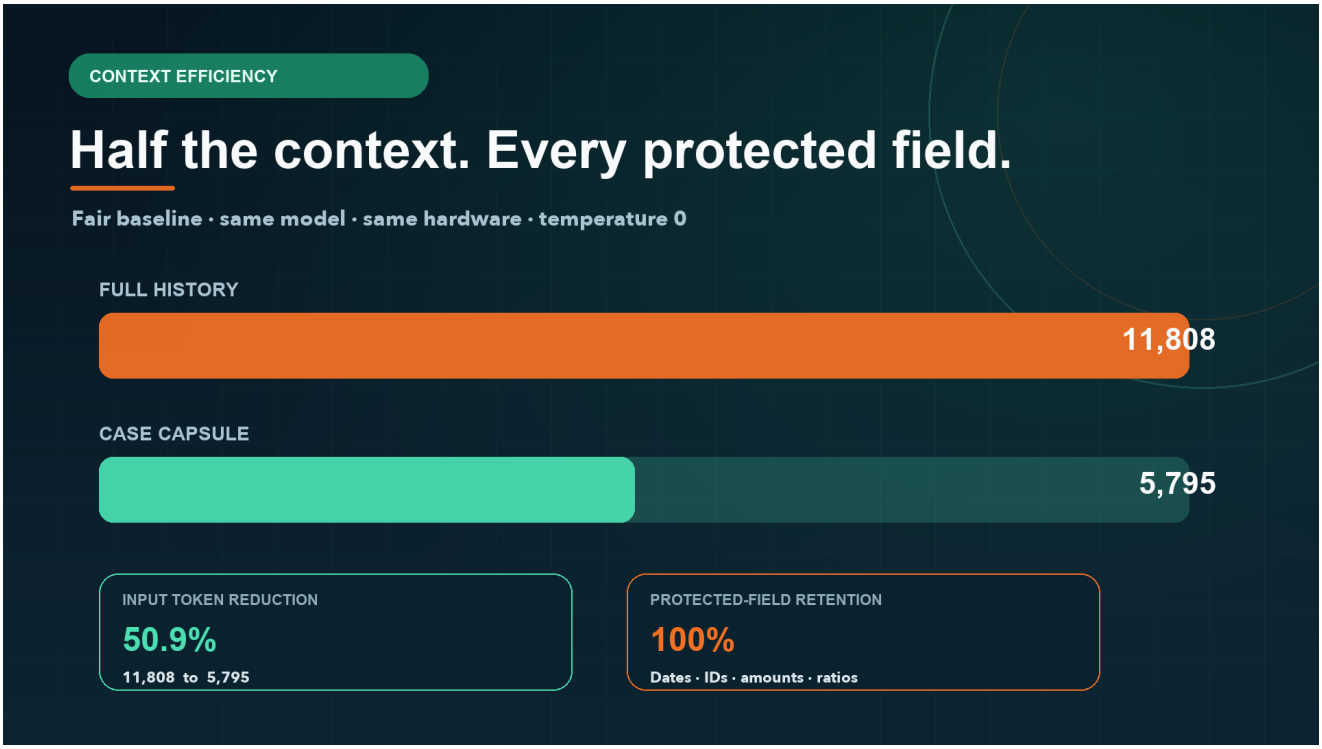


Figure 6. Case Capsule reduces context while retaining every protected field.

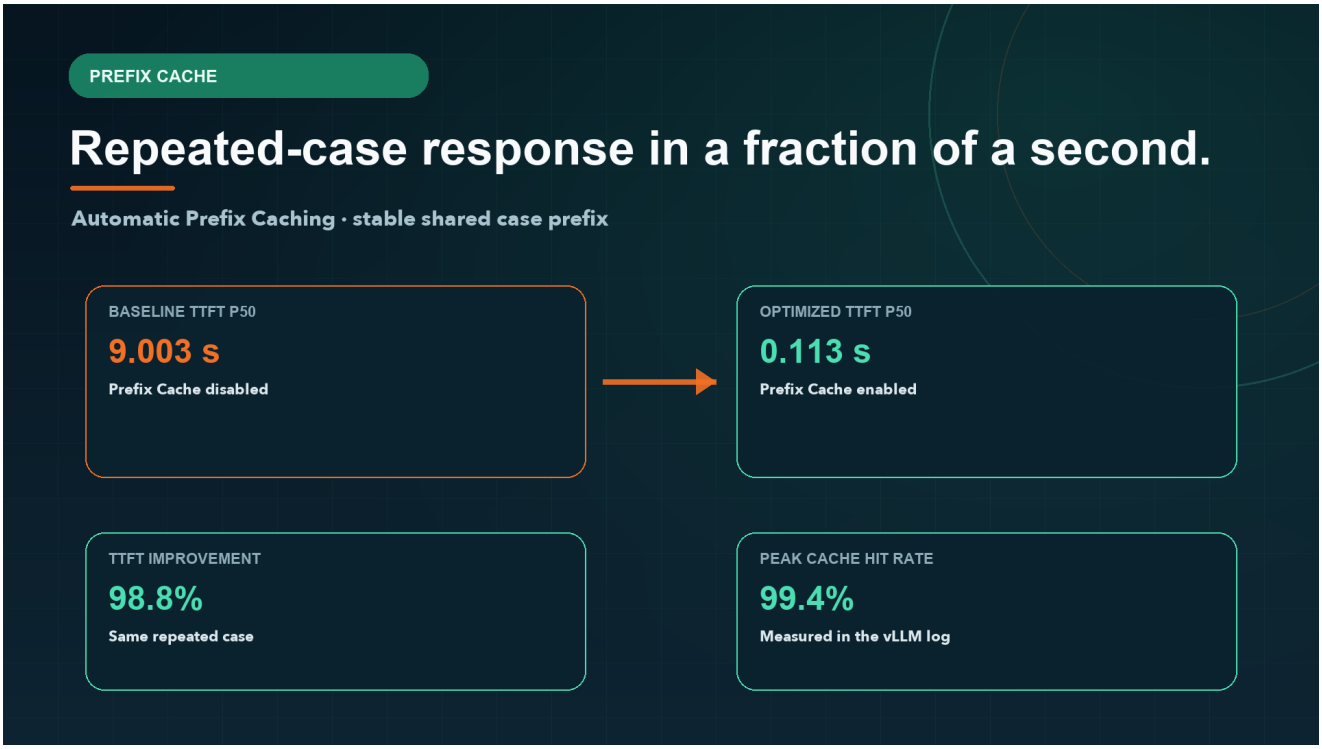


Figure 7. Automatic Prefix Caching accelerates the repeated-case first token.

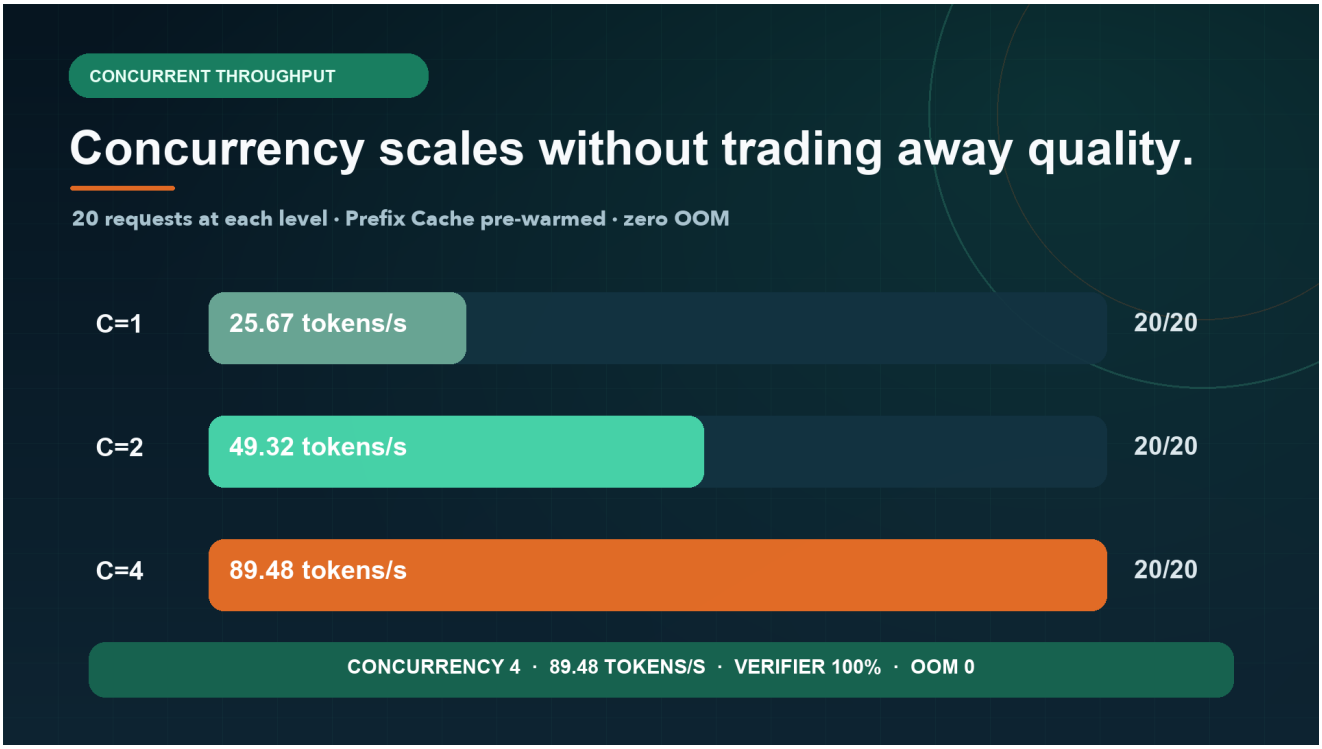


Figure 8. Concurrency 1, 2, and 4 each complete 20/20 requests.

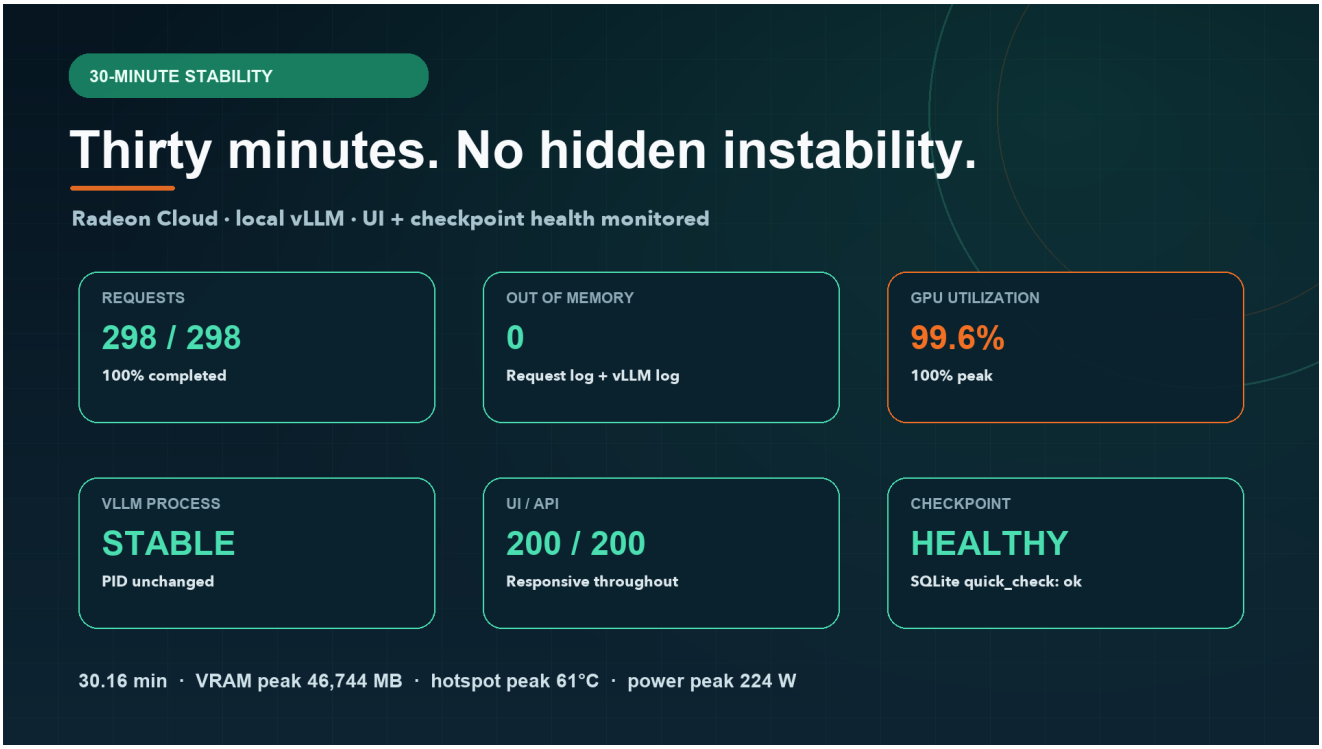


Figure 9. The 30-minute run completes 298/298 requests with zero OOM.

12. Deployment

Model inference, Agents, legal index, case files, checkpoints, and logs reside on the same Radeon Cloud instance. vLLM listens only on 127.0.0.1:8000; the UI listens on 127.0.0.1:7860. A private SSH tunnel is used for remote demonstration, and no public model port is exposed.

The validated remote evaluation path is:

Mac Chrome (127.0.0.1:7860) -> SSH Tunnel -> Radeon Cloud Gradio (127.0.0.1:7860) -> LangGraph Agent -> local vLLM (127.0.0.1:8000) -> AMD Radeon GPU

Only -L 7860:127.0.0.1:7860 is required. The Radeon Jupyter interface is not used to render the UI, port 8000 is not forwarded, and no Cloud Proxy is part of the approved path.

Mock mode requires Python and locked dependencies. Local vLLM mode first runs the endpoint doctor and stops if the co-located model is unavailable. The README records installation, launch, Golden Case, testing, and recovery steps.



Figure 10. Deployment proof aligned with the final README: no Cloud Proxy and no forwarding of the vLLM port.

13. Future Commercialization

The first controlled pilots can target insurance-claim assistance and vehicle owner rights services within one jurisdiction and accident type, using synthetic or de-identified historical cases. The immediate value is not to replace liability determination. It is to reduce evidence-sorting effort, surface missing materials and deadlines, make responsibility and payment discussions explainable, and give a human reviewer a reproducible case record.

A credible insurer pilot should start with one team and one workflow. Baseline the current first-review time, evidence-gap rate, rework, missed-window events, and handoff volume; run redacted cases through the Agent while retaining the insurer's existing approval roles; then compare operational results before any wider deployment. Adoption is earned with measured savings and auditability, not with an accuracy claim that the evidence does not support.

Production deployment requires tenant isolation, identity and RBAC, encryption and KMS, queueing and rate limits, metrics and alerts, model/rule version governance, backup and recovery, malicious-file sandboxing, and jurisdictional rule packages. Human authority over every external action remains a permanent design requirement.



Figure 11. Controlled pilot: redacted cases, existing approval gates, and measured savings.

ADOPTION PRINCIPLE	**Earn expansion with workflow evidence.** Keep approval rules in place and measure first-review time, gaps, rework, and missed windows.

Evidence and Limitations

All performance values are taken from Radeon Cloud JSON/CSV under benchmarks/results/. Functional results come from tests/ and the 2026-08-03 release-candidate rerun. Demo materials are synthetic. The project does not provide legal advice or guarantee review, litigation, appraisal, or insurance outcomes.

See amd_optimization.md for the complete parameter evidence, privacy_security.md for the local-data boundary, and model_license.md for model and license provenance.

Sources and Evidence

- Official competition repository and Track 2 rules:
  https://github.com/AMD-DEV-CONTEST/Radeon-hackathon-2026-07
- Radeon measurements: benchmarks/results/ baseline, optimized, concurrency, stability, and security evidence.
- Feature and E2E evidence: tests/, docs/ui_e2e_execution_report.md, and release_check/freeze_audit.md.
- Primary model: Qwen/Qwen3-VL-8B-Instruct; source code license: Apache License 2.0.
- Final demonstration:
  outputs/video_edit/Accident_Rights_Agent_Track2_DataLink_Demo_MAC_COMPAT_V4.mp4.

BOUNDARY	**Decision support-not legal advice.** Synthetic Demo results do not guarantee appraisal, review, litigation, or insurance outcomes.